

Deep Assessment Report

Tokenized Commercial Paper Issuance & Lifecycle — United States

A case-specific assessment report for tokenization viability and value creation.

September 22, 2026

A TVAC assessment by Astoval

CLARITY
BEFORE
COMMITMENT.

About this report

TVAC evaluates whether tokenizing a specific asset / structure creates net added value, using the five-factor model:

$$\text{Added Value} = (\text{New Opportunities (NO)} + \text{Cost Savings (CS)} + \text{Risk Reduction (RR)}) - (\text{Tokenization Costs (TC)} + \text{New Risks (NR)})$$

Where this report fits

TVAC is designed for the very early decision stage of a tokenization initiative. It tests whether the proposed use of tokenization appears likely to create genuine added value before significant resources are committed to legal structuring, issuance-platform or provider selection, detailed technical architecture, or implementation. A positive verdict is therefore not implementation approval; it indicates that the case may justify deeper legal, commercial, technical and operational work.

Case description summary

The initiating organization is a US-based corporate issuer with an existing, recurring commercial paper (CP) program used for short-term working capital. The proposed design seeks to introduce a tokenized CP note format (economically equivalent to our existing CP) for professional and accredited investors only. Today the program relies on multiple intermediaries, manual confirmations, repeated reconciliations between the issuer's treasury team, issuing/placing agents, custodians, and investors, and a settlement cycle that creates operational friction and occasional settlement fails. The scope is deliberately conservative: a single US issuance + distribution perimeter, whitelisted participation, and institutional-grade custody.

The full user-supplied case description and structured input record are reproduced in Appendix E.

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Methodology	TVAC — Tokenization Value-Add Calculator
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01 | Executive summary

TOKENIZATION VALUE VERDICT

Viable

ADDED VALUE

+9.00

plus 16.00 minus 7.00

Headline: The analysis supports a viable case for proceeding with tokenization under the proposed direction. The assessment lands at Viable with an estimated Net Added Value of 9.00 (Plus: 16.00 · Minus: 7.00).

Important interpretation note: This verdict concerns tokenization added value only. A Viable or Conditional Go verdict does not mean that the underlying asset is a good investment, that projected returns are realistic, that credit risk is acceptable, or that the instrument is suitable for any investor.

Yield neutrality note: The stated coupon, yield, interest rate, or expected return has not been treated as tokenization value in itself. TVAC does not assess whether projected returns are realistic, sustainable, appropriately priced for risk, or suitable for investors. Yield may be relevant to investor demand, but it does not by itself demonstrate that tokenization adds value.

Case design stage: Structured Concept

This assessment is based on a partially defined design. The case is sufficiently clear for a meaningful case-specific evaluation, but several structural choices remain open.

1.1 What is different about this case

What is different about this case is that the issuer already has a conventional commercial-paper programme and is testing whether tokenization can improve issuance, settlement, position reporting and lifecycle administration. The value case therefore depends on measurable process improvement, authoritative records, DvP and recovery controls, and retirement of legacy steps — not on new capital access, anchor investors, market making or assumed secondary liquidity.

KEY TAKEAWAYS

- 01 Key findings:** The user describes recurring manual work in the current CP program, including repeated reconciliations and exception handling across treasury, agents, custodians, and investors, which gives tokenization a concrete cost baseline to attack. Across New Opportunities, Cost Savings and Risk Reduction, the combined Plus score is 16.00.
- 02 Key constraints:** This design still carries material implementation burden through custody, issuance and recordkeeping infrastructure, legal and compliance work, and operating-model change across a live commercial paper program. Across Tokenization Costs and New Risks, the combined Minus score is 7.00.
- 03 Decision posture:** Proceed with disciplined implementation. Preserve the features that create value, but avoid unnecessary expansion of jurisdictional scope, liquidity promises, or technical complexity beyond what the case actually needs.
- 04 Impact:** This is primarily a controlled progression case. The upside appears plausible under the stated design, but it must be validated and captured without reintroducing avoidable TC/NR through overengineering or distribution overreach.

02 | Asset class & market context

This is primarily an issuance-and-lifecycle optimisation case within an established recurring debt-issuance programme. Tokenization value depends on whether the tokenized tranche can improve allocation, settlement, authoritative position records, reporting, interest or discount and maturity administration, exception handling and reuse across later issuances. It is not assessed as a new-capital-access or secondary-liquidity strategy.

Current market phase: Case-derived structural context only; external market maturity is not assessed from verified market sources for this case.

What this context means for the case

Against this market backdrop, the points below show which features of the proposed structure support the value case and which still constrain it.

Structural factors supporting the case:

- The issuer has an established conventional issuance-programme baseline against which operational improvement can be tested.
- The stated pilot targets concrete issuance and servicing frictions: allocation, settlement, position reporting, interest or discount administration and maturity processing.
- The professional-investor and no-liquidity-thesis perimeter keeps the pilot focused on controlled operational value rather than speculative market expansion.

Structural factors that still constrain the case:

- The authoritative record, custody or wallet-control model and responsibility split across issuer, arranger, registry operator and settlement bank remain unresolved.
- Cash-leg finality, failed-settlement handling, cancellation, rollback and recovery must be validated before settlement-sensitive benefits can be credited.
- A smaller tranche may struggle to absorb fixed setup costs, and value weakens materially if legacy issuance and servicing steps remain permanently in parallel.

03 | Analysis result

ADDED VALUE

+9.00

PLUS

16.00

MINUS

7.00

Verdict rationale:

This case is assessed as Viable because the current user-defined structure produces Added Value +9 (plus 16 minus 7), which places the case in the Viable band under the TVAC methodology. In practical terms, the strongest supportive driver is Cost Savings (6/10), while the main friction is Tokenization Costs (5/10). The user describes recurring manual work in the current CP program, including repeated reconciliations and exception handling across treasury, agents, custodians, and investors, which gives tokenization a concrete cost baseline to attack. This design still carries material implementation burden through custody, issuance and recordkeeping infrastructure, legal and compliance work, and operating-model change across a live commercial paper program. This rationale concerns tokenization added value only and does not validate legal, regulatory, tax, financial, investment, credit, yield, issuer, or implementation suitability.

Factor score profile

Factor	Score	Case-specific driver
New Opportunities	5/10	The main new opportunity is not broader distribution or liquidity; it is the ability to redesign a recurring U.S. commercial paper workflow around a shared position record and tighter transfer controls.
Cost Savings	6/10	The user describes recurring manual work in the current CP program, including repeated reconciliations and exception handling across treasury, agents, custodians, and investors, which gives tokenization a concrete cost baseline to attack.
Risk Reduction	5/10	The described model is intended to reduce reconciliation breaks and settlement fails by concentrating records and lifecycle events into a more controlled operating process.
Tokenization Costs	5/10	This design still carries material implementation burden through custody, issuance and recordkeeping infrastructure, legal and compliance work, and operating-model change across a live commercial paper program.
New Risks	2/10	New risk creation is relatively limited because the case is U.S.-only, excludes retail investors, does not rely on a secondary-liquidity promise, and uses a controlled OTC and custody model.

Added Value: 9.00 (plus 16.00 / minus 7.00)

What still needs to be verified:

These are the main points that should be independently confirmed before moving from concept toward implementation.

- No verified case-specific regulatory claim is activated; approved baseline context supports structured scoping, while case-specific legal application remains unresolved.
- Obtain written legal and control analysis of the token layer, authoritative records, participant and intermediary roles, custody or collateral control, settlement finality and the operating perimeter.
- Document and validate the relationship between the tokenized workflow, authoritative custody or agent records, collateral-control rights, asset- and cash-leg finality, exception handling, rollback and recovery.

Regulatory status: United States · Case-specific applicability unverified

TVAC has identified relevant regulatory context for the stated issuance jurisdiction and case-specific legal questions, but it has not verified how the law applies to the final structure. Final legal treatment should therefore be confirmed by qualified local counsel before implementation.

Sources used and outstanding validation questions are summarized in Appendix A.5.

The score becomes useful only when the structure behind it is coherent. The sections below test the value pathways, participant or investor perimeter, operating burden, regulatory path, custody and transfer fit, and—where material—the settlement and cash-leg design that must support the claimed benefits.

3.1 Value pathways

This section isolates the mechanisms through which tokenization could create incremental value over a realistic conventional alternative. It is not an assessment of whether the underlying asset or financing need is attractive in its own right.

- The primary value logic is issuance-lifecycle optimization for an existing U.S. commercial paper program, not capital formation expansion and not a liquidity-led STO thesis.
- The user describes a recurring CP workflow with multiple intermediaries, manual confirmations, repeated reconciliation steps, and occasional settlement fails; tokenization is intended to compress these operational touchpoints around a shared position record.
- Because the proposed tokenized CP is economically equivalent to the existing CP and is aimed at professional and accredited investors only, the value case rests mainly on process redesign, control improvement, and exception reduction rather than on new investor rights.
- The stated commitment to decommission duplicated legacy steps for the tokenized tranche is important: if implemented as described, it supports structural rather than merely cosmetic efficiency gains.

3.2 Investor segmentation

Investor scope affects both upside and friction. A narrower professional or institutional audience can reduce distribution and compliance complexity, but it can also limit the access or liquidity benefit that tokenization is expected to create.

- The investor perimeter is intentionally narrow: professional and accredited investors only, with retail excluded. That supports operational control and narrows the onboarding and communications surface.
- For this U.S. CP case, investor segmentation matters less as a demand-expansion story and more as a compliance and processing discipline: only holders who can operate inside whitelist, custody, and transfer-approval rules should be included.
- Because the case does not pursue public exchange access or retail distribution, the investor proposition must be practical: easier confirmations, cleaner post-trade processing, and predictable maturity handling.

3.3 Operational costs

This section distinguishes plausible process savings from the additional cost of introducing the token layer. A saving becomes a strong value driver only when it can be compared with the current process and is not offset by data, integration, provider or compliance work.

- The user expects lower agent and custody operations per issuance and fewer manual exceptions, but no quantified baseline is provided, so TVAC treats the savings case as plausible rather than measured.
- The strongest cost-saving mechanism is the stated removal of duplicated legacy steps for the tokenized tranche. Without that retirement, tokenization would risk becoming an added layer rather than a simpler operating model.
- Material cost categories are still likely to include custody, issuance and registry tooling, legal structuring, operational change management, and ongoing governance around permissions and records.
- Because this is an existing recurring U.S. CP program rather than a new fundraising product, even modest per-issuance efficiencies can accumulate if the tokenized workflow is adopted repeatedly.

3.4 Regulatory path

This is regulatory scoping, not a legal opinion. The purpose is to identify the verified regulatory context and the design questions that materially affect the case, while making clear what still requires qualified jurisdiction-specific validation.

- TVAC has approved baseline regulatory context for United States, but it has not activated a verified case-specific local claim. The frameworks below are used for structured scoping only; applicability and current detailed conditions remain to be independently confirmed.
- At scoping level, a token representing debt, equity, fund interests or another investment exposure should be analysed under the U.S. federal securities-law framework. Tokenization does not itself determine the instrument's legal classification.
- For U.S. securities capital formation, Regulation D, Regulation A and registered offerings are distinct pathways. Regulation S may be relevant to offshore offers and sales. TVAC may compare their possible relevance, but must not conclude that the final case qualifies for any pathway.
- Broker-dealer, alternative trading system, transfer-agent, custody and recordkeeping questions may arise according to the functions actually performed, the instrument classification and the proposed secondary-transfer model. A technology label does not decide those roles.

3.5 Custody & venue fit

Custody and venue design should be proportionate to the value thesis. If the case does not depend on continuous secondary trading, extra custody, venue or transfer complexity

should be justified by a concrete operational, control or distribution benefit.

- A regulated-custodian model is directionally aligned with a conservative U.S. commercial paper design aimed at professional and accredited investors rather than retail token holders.
- The OTC venue choice and explicit absence of a public exchange initiative fit the stated objective of workflow efficiency, controlled transfers, and limited secondary activity.
- The key fit question is not venue breadth but whether custody, transfer approvals, and recordkeeping can support the issuer's goal of reducing manual exceptions rather than recreating them in a new form.

3.6 Settlement & Cash-Leg Readiness

Settlement is assessed against the case's actual value logic, not against an assumption that tokenization requires digital money or T+0. The key question is whether the cash leg can operate reliably for the intended investors and operating model without unnecessary complexity.

- Why it matters: The claimed value depends directly on settlement-sensitive features such as DvP, fewer settlement fails and tighter transfer controls. The tokenized CP leg and the corresponding cash movement therefore need to work together reliably.
- Current position: the stated payment / settlement model is T+1/T+2 securities settlement. Its readiness is Defined conventional design. This is a design/readiness assessment, not evidence that settlement finality has already been achieved.
- What must be proven: asset- and cash-leg finality, synchronization, failed-settlement handling, exception management and recovery. Until those points are evidenced, settlement-dependent benefits should remain conditional rather than treated as realized outcomes.
- Assessment basis: this treatment reflects the stated asset/instrument type (wrapper) and liquidity intent (none); TVAC does not assume that every tokenized asset requires advanced digital money or T+0 settlement.

04 | Stakeholder value & alignment

This chapter asks whether the parties needed to make the structure work are economically and operationally aligned. It focuses on the value each party receives, the burden and responsibility it carries, and the dependencies that could weaken the project's overall Added Value.

4.1 Value & alignment overview

This view does not re-score the five TVAC factors for each party. It shows instead what each stakeholder stands to gain, what burden or responsibility it carries, and what must be aligned for the project value case to hold.

Issuer / programme sponsor

Expected value depends mainly on whether the tokenized tranche improves issuance, settlement, position reporting and lifecycle administration relative to the issuer's established recurring issuance programme, without creating a permanent parallel process.

Value received

Potential value comes from fewer manual handoffs, cleaner allocation and settlement, better position visibility and more controlled interest, maturity and lifecycle administration.

Cost, risk & responsibility

The issuer still carries legal-structure, integration, provider, governance and change-management costs before operational gains are demonstrated.

Alignment test

Define the authoritative record, responsibility split, lifecycle event model and evidence required to retire legacy steps.

Professional investors

Investor value depends mainly on clear rights, reliable position records, controlled settlement and accurate lifecycle servicing rather than on promised liquidity or expanded access.

Value received

Cleaner confirmations, position reporting and lifecycle communications may improve the holding and servicing experience.

Cost, risk & responsibility

Investors need clarity on the authoritative record, custody or wallet control, transfer restrictions, failed-settlement handling and fallback arrangements.

Alignment test

Require a clear rights-and-operations note covering record authority, settlement finality, custody, transfers, interest or discount, maturity processing and incident recovery.

Issuance, registry, custody and settlement providers

Provider value depends on delivering a coherent issuance and lifecycle stack that integrates with the issuer's existing systems and removes measurable friction rather than merely adding another technology layer.

Value received

Providers can create durable value through reliable registry integration, controlled DvP, lifecycle automation, audit evidence and reusable issuance workflows.

Cost, risk & responsibility

Responsibility gaps between arranger, registry or transfer operator, custodian, wallet provider and settlement bank can create high error and recovery costs.

Alignment test

Agree a detailed RACI, authoritative-record model, DvP sequence, exception authority, recovery process, service levels and exit or portability provisions.

4.2 Issuer / programme sponsor

Value received

- Potential value comes from fewer manual handoffs, cleaner allocation and settlement, better position visibility and more controlled interest, maturity and lifecycle administration.
- A successful pilot can create reusable operating capability for later issuances without relying on broader investor access or secondary-market liquidity.

Costs, risks & responsibilities

- The issuer still carries legal-structure, integration, provider, governance and change-management costs before operational gains are demonstrated.
- Value erodes if the tokenized tranche remains dependent on duplicate records, manual approvals or permanent parallel servicing.

Alignment & actions

- Define the authoritative record, responsibility split, lifecycle event model and evidence required to retire legacy steps.
- Compare the full tokenized-tranche cost and operating performance with the existing issuance-programme baseline before scaling.

4.3 Professional investors

Value received

- Cleaner confirmations, position reporting and lifecycle communications may improve the holding and servicing experience.
- A validated DvP process can reduce settlement uncertainty without changing the conventional economic terms of the instrument.

Costs, risks & responsibilities

- Investors need clarity on the authoritative record, custody or wallet control, transfer restrictions, failed-settlement handling and fallback arrangements.
- A new token layer adds provider and operational dependencies even where no public secondary market is planned.

Alignment & actions

- Require a clear rights-and-operations note covering record authority, settlement finality, custody, transfers, interest or discount, maturity processing and incident recovery.
- Do not treat the pilot as offering additional liquidity unless a separate, validated transfer pathway is actually introduced.

4.4 Issuance, registry, custody and settlement providers

Value received

- Providers can create durable value through reliable registry integration, controlled DvP, lifecycle automation, audit evidence and reusable issuance workflows.
- A repeat issuer offers a credible path to reuse if the first tranche proves operationally superior to the conventional process.

Costs, risks & responsibilities

- Responsibility gaps between arranger, registry or transfer operator, custodian, wallet provider and settlement bank can create high error and recovery costs.
- Commercial value is weak if fees and integration burden are not tied to measurable process retirement and service outcomes.

Alignment & actions

- Agree a detailed RACI, authoritative-record model, DvP sequence, exception authority, recovery process, service levels and exit or portability provisions.
- Tie provider commitments and scale-up to evidenced reductions in manual work, reconciliation and lifecycle-processing burden.

4.5 Who bears the costs and risks?

For this qualified high-volume institutional workflow, the initiating institution may carry most front-loaded integration and governance costs. That does not by itself mean the economics favour service providers. The relevant test is whether the initiating institution captures measurable cumulative savings and risk reduction once the workflow reaches run-state.

- A neutral or low near-term initiating-institution position should trigger an evidence and cost-allocation review, not an automatic rejection. It becomes a warning sign where the expected operational benefits are not demonstrated, provider fees absorb the realised gains, or key service obligations remain misaligned with value capture.
- Eligible counterparties should still receive clear evidence of operational controls, custody synchronisation, settlement finality and the allocation of exception-handling costs.
- Where operational benefits are evidenced and all stakeholders are positive, focus on disciplined scale-up, service continuity and ongoing measurement of realised value.

05 | Factor breakdown – detailed explanation

This chapter explains the causal basis of the five scores: which facts, assumptions and evidence gaps create upside, and which ones introduce additional friction. The three positive factors ask what tokenization adds relative to a realistic conventional alternative; the two negative factors ask what extra cost or risk the token layer introduces. A low positive score therefore means limited or weakly evidenced tokenization benefit, while a low negative score means limited additional friction. The scores are calibrated methodology judgments, not measured outcomes or legal conclusions.

$$\text{Added Value} = (\text{New Opportunities} + \text{Cost Savings} + \text{Risk Reduction}) - (\text{Tokenization Costs} + \text{New Risks})$$

The factor profile shows a positive tokenization value case. The strongest upside comes from cost savings, while the principal friction is tokenization costs; the explanations below set out the case-specific mechanisms and evidence gaps behind those judgments. With an Added Value score of 9, the value case is clearly positive rather than marginal.

New opportunities (NO)

- The main new opportunity is not broader distribution or liquidity; it is the ability to redesign a recurring U.S. commercial paper workflow around a shared position record and tighter transfer controls.
- Automated issuance confirmations, accrual or coupon reporting, and maturity repayment notices create a token-enabled operating model that is harder to replicate cleanly across multiple parties with disconnected records.
- The upside remains bounded because the case is intentionally U.S.-only, professional and accredited only, and explicitly not a public exchange initiative.

Cost savings (CS)

- The user describes recurring manual work in the current CP program, including repeated reconciliations and exception handling across treasury, agents, custodians, and investors, which gives tokenization a concrete cost baseline to attack.
- The strongest support for savings is the explicit plan to decommission duplicated legacy steps for the tokenized tranche rather than maintain permanent double-booking.
- Savings are still contingent on adoption by operating parties and on actual process retirement, but the recurring nature of the CP program makes per-issuance efficiency gains more valuable than in a one-off issuance.

Risk reduction (RR)

- The described model is intended to reduce reconciliation breaks and settlement fails by concentrating records and lifecycle events into a more controlled operating process.

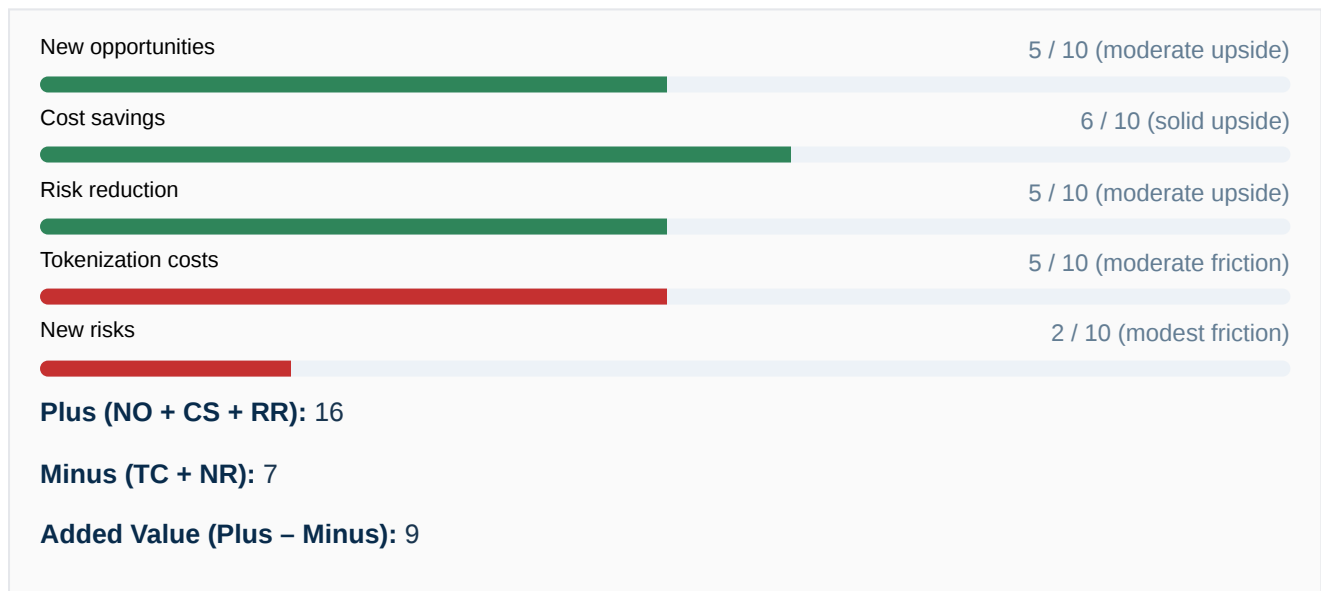
- Whitelist controls, issuer or agent-approved transfers, and institutional custody are aligned with a conservative non-retail U.S. CP perimeter and may support cleaner control over holder eligibility and movement.
- The risk-reduction case is strongest in operational and audit-trail terms; it still requires validation of authoritative records, DvP-style exception handling, and the actual control allocation among parties.

Tokenization costs (TC)

- This design still carries material implementation burden through custody, issuance and recordkeeping infrastructure, legal and compliance work, and operating-model change across a live commercial paper program.
- The need to coordinate issuer treasury, agents, custodians, and investors means integration and governance work are meaningful even though the scope is deliberately narrow.
- Costs appear manageable because the case avoids retail distribution, public exchange ambitions, and multi-jurisdiction expansion in the initial phase.

New risks (NR)

- New risk creation is relatively limited because the case is U.S.-only, excludes retail investors, does not rely on a secondary-liquidity promise, and uses a controlled OTC and custody model.
- The main residual risk categories are legal-technical mapping, provider dependency, role-mapping uncertainty, and the possibility that DvP-style controls do not work as intended with the stated T+1/2 cash rails.
- These are real implementation risks, but they are narrower than in a cross-border, exchange-traded, or retail-facing tokenization design.



What could change this verdict

These are the highest-impact conditions that could materially strengthen or weaken the current assessment. They are practical decision gates, not predictions.

- Distribution scope and investor segmentation (jurisdiction count, retail vs professional posture)
- Rights mapping and structural clarity (asset/SPV structure, cash-flow logic, investor protections)
- Custody / controls / transfer restrictions (institutional-grade requirements vs complexity)
- Liquidity assumptions (timing, venue feasibility, realistic market depth)
- Payment / cash-leg readiness where the case depends on DvP, digital cash, 24/7 settlement, collateral mobility or automated lifecycle payments
- Tokenization cost profile (legal/compliance/ops overhead and non-scalable components)

06 | Key risks & mitigation

This matrix prioritizes the report's identified risks by impact and likelihood. It is not a measured probability model; each position should be read together with the case-specific description and mitigation.

	1 Rare	2 Unlikely	3 Possible	4 Likely	5 Almost certain
5 High			R2 R3		
4 Med-high		R5	R1 R4 R6		
3 Medium	IMPACT ↑				
2 Med-low					
1 Low					

LIKELIHOOD →

Risk zones are indicative prioritization bands, not measured probabilities.

PRIORITY RISKS

R1 Misaligned incentives / vendor-capture risk

Impact 4 · Likelihood 3

Mitigation: Use stage-gating based on legal/control validation and measurable operational-readiness evidence; link provider fees and scale-up to issuance completion, process retirement, service levels, reconciliation performance and lifecycle-delivery outcomes; preserve portability and exit rights if the provider arrangement changes.

R2 Authoritative record ambiguity

Impact 5 · Likelihood 3

Mitigation: Obtain documented mapping between token records, custodian records, conventional documentation, and the legally authoritative ownership and transfer record.

R3 Parallel-process persistence

Impact 5 · Likelihood 3

Mitigation: Define and govern a formal retirement plan for duplicated confirmations and reconciliation steps for the tokenized tranche.

R4 Cash-leg and DvP mismatch

Impact 4 · Likelihood 3

Mitigation: Document the exact cash-movement workflow, settlement cut-offs, fail handling, and exception escalation for the T+1/2 payment model.

R5 Role-mapping and intermediary scoping risk

Impact 4 · Likelihood 2

Mitigation: Obtain independent U.S. legal analysis of the actual functions performed by issuer, agents, custodians, and any OTC transfer participants.

R6 Provider dependency and portability risk

Impact 4 · Likelihood 3

Mitigation: Negotiate binding service scope, pricing, data portability, operational service levels, and exit rights with core providers.

07 | Recommendations & next steps

7.1 Design priorities and decision gates

This is an issuance and lifecycle-optimisation case for a conventional recurring debt-issuance programme, not a capital-access or secondary-liquidity case. The priorities focus on proving operational improvement against the existing issuance baseline.

7.1.1 Prove improvement against the existing issuance-programme baseline

Measure manual handoffs, allocation and settlement effort, reconciliation, position reporting, interest or discount administration, maturity processing, exceptions and lifecycle-processing cost before the pilot. Scale only where the tokenized tranche produces a durable improvement over the established process.

7.1.2 Resolve authoritative records, responsibility and DvP recovery

Define the authoritative instrument and investor record, the role of the token layer, the responsibility split across issuer, arranger, registry or transfer operator, custodian, wallet provider and settlement bank, and the rules for finality, cancellation, reversal, rollback and recovery.

7.1.3 Retire legacy steps and align provider economics with evidence

Specify which legacy issuance and servicing steps may be removed after validation. Tie provider commitments, fees and scale-up to measurable process retirement, control evidence, service levels, portability and incident performance.

7.1.4 Execution implication for an issuance and lifecycle pilot

Proceed through a bounded professional-investor pilot only if the operating model, authoritative record, DvP and cash-leg controls, lifecycle servicing and recovery arrangements are independently validated. Do not add anchor-investor, fundraising, market-making or liquidity objectives unless the case is separately redesigned and reassessed.

7.2 Design what-if scenarios

These scenarios test the operational value of the tokenized tranche relative to the issuer's existing issuance programme; they do not assume new capital access or secondary liquidity.

- The pilot removes defined manual and reconciliation steps while preserving authoritative records and controls: Cost Savings and Risk Reduction become more durable.
- The token layer remains parallel to existing records and servicing: Tokenization Costs rise and the operational value case weakens.

- Cash-leg finality, failed-settlement recovery or responsibility allocation remains unresolved: New Risks should be treated as higher and scale-up should pause.

7.3 Suggested next steps

- **Establish the conventional issuance and servicing baseline and pilot success thresholds.**

A before-and-after baseline is needed to show whether tokenization actually improves the current process rather than merely changing the technology stack.

- **Document the authoritative record, token role, RACI, lifecycle event model and recordkeeping obligations.**

The assessment can identify the legal question, but implementation should not proceed until the relevant rights, roles and legal effect are confirmed for the final structure.

- **Validate DvP, cash-leg finality, failed-settlement, cancellation, rollback and recovery procedures end to end.**

The claimed operating benefit depends on both sides of settlement working together under normal and failed conditions, so finality and recovery must be demonstrated rather than assumed.

- **Agree which legacy steps can be retired and tie provider economics and scale-up to measured evidence.**

Staged commercial terms limit lock-in and keep provider economics tied to demonstrated operational value; audit, service-level, portability and exit rights preserve control if delivery underperforms.

Appendices

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Appendix A — Method & interpretation

A.1 How to use this appendix

This appendix explains how to interpret the assessment, what evidence is still needed, and which conditions could strengthen or weaken the case. It is not a technical manual for TVAC.

A.2 What the assessment establishes — and what it does not

This report provides informational decision support only. It does not constitute legal, regulatory, tax, financial, or investment advice, and it does not provide a compliance determination, valuation, or provider endorsement.

Case snapshot (as stated)

- Issuance jurisdiction: United States
- Sale jurisdictions / target markets: United States
- Asset / instrument type (indicative): wrapper
- Liquidity intent (as stated): none
- Payment / cash-leg indication: T+1/T+2 securities settlement
- Cash-leg dependency / readiness: Critical dependency / Defined conventional design readiness

What the Tokenization Value Verdict means

- Viable: Tokenization added value is plausible under the stated assumptions, subject to normal verification and execution.
- Conditional Go: Tokenization added value is design-dependent. Proceeding without meeting the listed conditions materially increases execution risk.
- Non-viable: Under current assumptions, tokenization is unlikely to outperform a conventional structure on a risk-adjusted basis.

What TVAC did not assess

- No definitive legal, regulatory, tax, or investment conclusions
- No asset valuation, yield projections, or market-price forecasts
- No assurance of secondary-market liquidity or investor demand
- No operational readiness audit beyond the indicative checklist

A.3 Minimum evidence before the next major commitment

Before proceeding toward implementation, decision-makers should be able to substantiate a minimum level of evidence across the areas below. This checklist is not exhaustive and does not imply regulatory sufficiency. It reflects common governance expectations in institutional-grade projects.

Highest-priority gaps in this case

(derived from inputs/completeness)

1. No verified case-specific regulatory claim is activated; approved baseline context supports structured scoping, while case-specific legal application remains unresolved.
2. Obtain written legal and control analysis of the token layer, authoritative records, participant and intermediary roles, custody or collateral control, settlement finality and the operating perimeter.
3. Document and validate the relationship between the tokenized workflow, authoritative custody or agent records, collateral-control rights, asset- and cash-leg finality, exception handling, rollback and recovery.

Structural & economic foundations

- Clear description of the underlying asset and ownership structure
- Defined economic rights and cash-flow logic
- Draft term sheet or structural outline
- Clear rationale for tokenization vs alternatives

Regulatory & distribution posture

- Issuance jurisdiction clearly defined
- Explicit list of sale jurisdictions
- Documented investor type assumptions
- Initial regulatory scoping (high-level)

Token & control design

- High-level token design and rights mapping
- Transfer restriction and whitelist logic (conceptual)
- Corporate actions and investor reporting approach
- Governance for smart-contract or protocol changes (if relevant)

Infrastructure & counterparties

- Custody model (conceptual)
- Indicative partner shortlist (non-binding)
- Issuance, registry, and lifecycle approach
- Payment / cash-leg and settlement approach, proportionate to the case's actual value proposition
- Early assessment of secondary trading (if any)

Cost & operational realism

- High-level CAPEX and OPEX view
- Identification of major cost drivers
- Clear internal ownership of ongoing operations
- Understanding of scalable vs non-scalable costs Lack of clarity across several of these areas materially increases execution risk, even where the conceptual case appears attractive.

A.4 What could strengthen or weaken the case

- Upside driver: Issuance and lifecycle improvement (value depends on measurable reductions in manual work, reconciliation, reporting and servicing)
- Risk driver: New risks (most sensitive to authoritative-record mapping, cash-leg finality, recovery design and provider-role clarity)
- Cost driver: Tokenization costs (most sensitive to fixed pilot costs, integration burden, duplicate processes and provider scope)

High-impact decision levers (practical)

- Legacy-process retirement: identify which allocation, reconciliation, reporting, coupon and maturity steps can be removed after validation.
- Authoritative record and controls: define the legally and operationally authoritative record, the role of the token layer and the responsibility split across the operating stack.
- Settlement and recovery: validate DvP sequencing, cash-leg finality, failed-settlement handling, cancellation, reversal, rollback and recovery.
- Provider economics: tie fees and scale-up to service levels, measurable process improvement, portability and incident performance.
- Reuse across later issuances: improve economics by reusing validated integration, governance and lifecycle capabilities rather than rebuilding them for each tranche.

How to use this

Improve the operating evidence and control model before adding distribution, fundraising, trading-support or liquidity objectives. Those objectives would materially change the case and require a separate assessment.

A.5 Evidence & source transparency

Basis of assessment

The assessment is based on the customer's case description and structured inputs, the fixed TVAC methodology, and any activated verified external sources. User-supplied operational and commercial claims have not been independently verified unless explicitly stated.

External sources used

- SEC — Exempt Offerings overview · U.S. Securities and Exchange Commission · checked 2026-07-15
- SEC — Regulation D Offerings · U.S. Securities and Exchange Commission · checked 2026-07-15
- SEC — Regulation A · U.S. Securities and Exchange Commission · checked 2026-07-15
- eCFR — Regulation S: offers and sales outside the United States · U.S. Securities and Exchange Commission / Electronic Code of Federal Regulations · checked 2026-07-15
- SEC — Guide to Broker-Dealer Registration · U.S. Securities and Exchange Commission · checked 2026-07-15
- SEC — Alternative Trading Systems · U.S. Securities and Exchange Commission · checked 2026-07-15
- SEC — Transfer Agents · U.S. Securities and Exchange Commission · checked 2026-07-15
- New York Fed — Repo Market Structure and Monetary Policy Implementation (tokenized repo discussion) · Federal Reserve Bank of New York · checked 2026-09-14
- BIS Annual Economic Report 2025 — The next-generation monetary and financial system · Bank for International Settlements · checked 2026-09-14
- DTCC — U.S. trades successfully processed using DTC-tokenized assets · Depository Trust & Clearing Corporation · checked 2026-09-14

What remains unverified

- Which record is legally authoritative for the debt instrument and investor positions, and what legal and operational role does the token layer perform relative to that record?
- How are responsibilities divided among issuer, arranger, transfer or registry operator, custodian, wallet provider and settlement bank for issuance, position maintenance and lifecycle events?
- How are asset-leg and cash-leg finality, DvP sequencing, failed settlement, cancellation, reversal, rollback and recovery defined for the pilot?
- Which eligibility, onboarding, disclosure, communications, marketing and transfer-control requirements apply in each stated jurisdiction to the professional-investor perimeter?
- How must interest or discount, principal, maturity, corporate-action, tax and recordkeeping events be reflected across the authoritative record, token layer and existing issuer systems?
- Which legacy issuance and servicing steps may be retired after validation, and what evidence is required before parallel processing can be discontinued?

Appendix B — Operational roadmap

This roadmap translates the assessment into a staged implementation path for the specific case type. It is not a project plan or implementation instruction: each phase should be adapted to the final legal structure, counterparties, providers and evidence available. Progress only when the decision gate for the preceding phase has been met.

B.1 Operational roadmap (structured)

The items below are indicative planning scaffolding. They are not a substitute for jurisdiction-specific legal, tax, financial, and operational due diligence, and they must be tailored to the selected asset type, wrapper structure, investor segment, custody model, venue, and settlement design.

B.2 Phase 1 — Preparation & feasibility

Case-specific focus for this phase: Primary issuance jurisdiction: United States. Initial distribution scope: United States. Initial investor focus: professional investors, accredited / qualified investors.

Objectives

- Clarify the underlying asset/instrument, economic rights, and proposed token wrapper.
- Define target investor segments (professional / eligible / retail, if applicable) and distribution scope.
- Establish internal alignment on objectives, success criteria, and risk appetite.

Key decisions

- Token rights model (equity-like, debt-like, fund-like, revenue share, participation note, etc.).
- Transfer restriction regime (whitelisting, holding periods, jurisdictional gating).
- Custody and settlement direction (account-based vs. wallet-based; T+0 vs. T+1+).

Deliverables (checklist)

- One-page case definition: asset, wrapper, jurisdiction(s), investor segment, and target size.
- Preliminary term sheet outline (rights, restrictions, fees, lifecycle events, disclosures).
- High-level implementation estimate (time bands, cost bands, key dependencies).
- Initial risk register (top 10–15 risks with owners and first mitigations).

Primary stakeholders

- Issuer/asset owner (business sponsor, finance, legal/compliance, operations/IT).
- External legal counsel (securities, funds, structuring), tax adviser, and (if relevant) auditor.
- Shortlisted tokenization platform/technology provider, regulated custodian, and potential venue(s).

Dependencies / gates

- Clear legal characterization of the instrument and its regulatory perimeter in the issuance jurisdiction.
- Agreement on must-have constraints (investor eligibility, reporting, custody, settlement, data handling).
- Internal approval to proceed to structured design (budget and governance gate).

Risk hotspots

- Unclear instrument classification or incompatible distribution intent (e.g., retail scope without a verified pathway).
- Mismatch between user-stated transfer expectations and the legally and operationally supportable process.
- Hidden operational burdens (cap-table/admin, reporting, corporate actions, tax documentation).

B.3 Phase 2 — Legal, tax & regulatory structuring

Case-specific focus for this phase: Core legal and tax analysis should start from United States. Distribution permissions must be validated for United States. Generic TVAC disclosure label selected in the input: Exempt. The applicable local legal category remains to be confirmed. Transfer-control design to validate: Whitelist + accredited/professional-only + issuer/agent-approved transfers.

Objectives

- Obtain independent legal validation of a legally supportable issuance and distribution structure for the intended jurisdiction(s).
- Define disclosure, governance, and ongoing compliance obligations for the lifecycle of the instrument.
- Obtain qualified tax analysis of the proposed wrapper and cash-flow mechanics for issuer and investors.

Key decisions

- Issuance vehicle choice (direct issuer vs. SPV; fund wrapper vs. corporate wrapper).
- Offering and documentation pathway verified for the stated jurisdiction(s); do not infer a legal category from interface labels.
- Investor onboarding model (KYC/AML responsibilities, reliance model, delegated vs. in-house).
- Choice of legal documentation stack verified by qualified local advisers.

Deliverables (checklist)

- Legal characterization memo and jurisdiction matrix (requirements by country/investor segment).
- Draft offering documentation and investor disclosures.

- Transfer restriction policy and rulebook (whitelist logic, jurisdictional gating, and any explicitly selected holding-period rules).
- Tax memo(s) covering withholding, reporting, and cross-border distribution considerations.
- Compliance operating model (roles, escalation paths, record-keeping, audit trail).

Primary stakeholders

- Issuer legal/compliance, external counsel, tax advisers.
- Custodian/compliance function (if regulated custody is used), transfer agent (if applicable).
- Any transfer provider or environment explicitly retained in scope and independently validated.

Dependencies / gates

- Independent legal confirmation of the offering pathway and distribution permissions across the stated target jurisdictions.
- Operational feasibility of investor onboarding and transfer controls (including data privacy constraints).
- Formal sign-off on documentation pack and compliance model.

Risk hotspots

- Jurisdiction conflicts (marketing rules, suitability requirements, local registration triggers).
- Under-scoped ongoing obligations (periodic reporting, investor communications, corporate actions).
- Tax frictions that erode the added value thesis (withholding, reporting, complexity costs).

B.4 Phase 3 — Build, integration & vendor onboarding

Case-specific focus for this phase: The current custody design indicates regulated custodian. Its precise legal and operational role remains to be confirmed. The current transfer environment indicates controlled OTC transfers; this does not establish a local legal venue classification. The current settlement design indicates T+1/T+2 securities settlement. Its operational and legal fit remains to be validated. Transfer-rule implementation must reflect: Whitelist + accredited/professional-only + issuer/agent-approved transfers.

Objectives

- Finalize the technology stack and integrate custody, settlement, onboarding, and reporting workflows.
- Implement transfer rules and lifecycle event handling (issuance, redemptions, distributions, corporate actions).
- Establish operational readiness (controls, monitoring, incident response, reconciliation).

Key decisions

- Token standard / ledger choice and compatibility with custody and venue requirements.
- Settlement model (atomic delivery-versus-payment vs. coordinated settlement through intermediaries).
- Data and reporting architecture (who stores what; audit trail; privacy boundaries).
- Outsourcing vs. in-house responsibility split (transfer agent, admin, reporting).

Deliverables (checklist)

- Final technical architecture and integration plan (custody, venue, payments, identity/KYC).
- Rule engine / whitelist implementation and test suite (including jurisdictional gates).
- Operational runbooks (reconciliation, incident response, corporate actions, investor support).
- UAT results and sign-offs (issuer, custodian, venue, admin/TA as relevant).
- Security review (key management model, access control, change management, monitoring).

Primary stakeholders

- Issuer operations/IT, tokenization platform/tech provider, regulated custodian where explicitly selected.
- Selected transfer-environment integration team, payment rails (if applicable), admin/TA providers.
- Compliance and risk functions for control design and approval.

Dependencies / gates

- Signed vendor contracts, SLAs, and responsibility matrix (RACI) across the stack.
- Completed integration testing and reconciliation proof (end-to-end lifecycle flows).

- Readiness gate: controls, monitoring, and support operations in place.

Risk hotspots

- Integration complexity and reconciliation gaps across custody/venue/admin systems.
- Rule enforcement drift (transfer restrictions not consistently applied across venues/wallets).
- Operational single points of failure (provider concentration, key management, support load).

B.5 Phase 4 — Issuance, distribution & launch

Case-specific focus for this phase: Any legally required launch permissions and onboarding constraints must be independently confirmed for: United States. User-selected generic investor segment for launch planning: professional investors, accredited / qualified investors. Local legal category mapping remains to be confirmed. The launch transfer environment currently indicates controlled OTC transfers; this does not establish a local legal venue classification. The current liquidity intent is no secondary liquidity planned. No market status, matching demand or liquidity support is inferred.

Objectives

- Execute issuance mechanics and investor onboarding under the independently verified offering pathway.
- Operationalize disclosures, subscriptions, allocations, and post-issuance record-keeping.
- Operationalize only the user-stated and legally validated transfer process, if any; no venue listing, market making or liquidity support is assumed.

Key decisions

- Distribution channel mix (direct, intermediated, platform marketplace, placement agents) as legally validated.
- Subscription mechanics and settlement timing (funding rails, DVP, cutoff times) as actually specified.
- Transfer model, if any: stated approvals, eligibility, timing and limitations; no cadence or support mechanism is assumed.

Deliverables (checklist)

- Final investor communication pack and distribution checklist.
- Subscription and allocation runbook; reconciliation and exception handling.
- Go-live monitoring plan and incident escalation paths.
- Documented transfer-process setup, if applicable, and evidence-based monitoring metrics.

Primary stakeholders

- Issuer distribution team, platform operator, onboarding/KYC provider.
- Custodian, settlement and transfer providers explicitly selected for the stated process.
- Legal/compliance for marketing controls and ongoing disclosure obligations.

Dependencies / gates

- Approved marketing/distribution materials and compliance sign-off.
- End-to-end dry run of subscription, allocation, and post-issuance reporting.
- Operational readiness gate: support, monitoring, and governance in place.

Risk hotspots

- Distribution frictions (slow onboarding, suitability issues, cross-border marketing constraints).
- Unsupported transfer or liquidity expectations relative to the actual process and evidence.
- Operational load spikes (investor support, reconciliation exceptions, reporting deadlines).

B.6 Phase 5 — Post-issuance operations & monitoring

Case-specific focus for this phase: Ongoing governance and evidence trails must remain defensible under United States. Post-issuance compliance must continue to support holders across: United States. Ongoing operations must reflect the regulated custodian design once its precise role has been independently confirmed. Post-launch monitoring must reflect the actual validated transfer process currently described as controlled OTC transfers.

Objectives

- Run ongoing compliance, reporting, and investor communications through the instrument's lifecycle.

- Operate lifecycle events (distributions, redemptions, corporate actions) with strong controls and auditability.
- Monitor provider performance, costs, and risk posture; adjust as regulations and market conditions evolve.

Key decisions

- Governance cadence (board/committee oversight, change approvals, vendor reviews).
- Reporting scope and frequency (investor reporting, regulatory reporting, audit readiness).
- Contingency planning (provider exits, custody changes, venue strategy shifts).

Deliverables (checklist)

- Ongoing reporting calendar and control framework (incl. audits and reconciliations).
- Lifecycle event runbooks and evidence retention procedures.
- Vendor performance KPIs and incident log; periodic risk reviews.
- Contingency and change-management playbook (custody/venue migrations, rule updates).

Primary stakeholders

- Issuer operations, compliance, finance; custodian; admin/TA (where applicable).
- Transfer and settlement providers explicitly retained in scope, if any.
- External auditors/advisers for periodic reviews and regulatory updates.

Dependencies / gates

- Stable operating model and evidence trail sufficient for audit/regulatory scrutiny.
- Clear change-management process for token rules, disclosures, and provider configuration.
- Budgeting and resourcing for ongoing operations (often underestimated in early planning).

Risk hotspots

- Compliance drift as rules change or distribution expands across jurisdictions.
- Operational fragility under low-liquidity or stress scenarios (settlement failures, disputes).
- Provider concentration risk and cost creep relative to value delivered.

Appendix C — TVAC Provider Directory (illustrative)

This appendix is an illustrative reference layer intended to help the reader think about which types of external partners may be relevant in a tokenization project.

It is not a recommendation list, procurement shortlist, or ranking. Inclusion or omission does not imply endorsement, criticism, quality judgment, regulatory approval, or case suitability. Any provider mentioned here must be assessed independently for legal fit, jurisdictional coverage, operating model fit, technical capability, conflicts, cost, and execution reliability. This appendix is non-exhaustive and subject to change.

B1. How to use this appendix

Use this appendix to identify the provider categories a case may require, not as a signal that a specific provider should be engaged.

In many cases, the main value of the directory is not the names themselves, but the discipline of asking:

- Which functions must be covered?
- Which functions can be combined?
- Which functions must remain regulated or jurisdiction-specific?
- Which dependencies create concentration risk or vendor-capture risk?
- Which parts of the stack are critical before launch, and which can wait?

B2. Typical provider categories

Depending on the case, a tokenization project may require some or all of the following partner categories:

- legal & regulatory advisers
- tax advisers
- issuance / tokenization platform
- smart-contract / token engineering support
- custody / wallet infrastructure
- transfer agent / registry administration
- fund administration or SPV administration
- trading venue / ATS / MTF / exchange partner
- transfer or liquidity-support provider, only where explicitly required and legally permitted
- KYC / AML / onboarding provider
- payment / settlement / banking partner
- audit / assurance / cybersecurity reviewers
- valuation, data, or oracle-related providers
- distribution / placement / investor onboarding partners
- systems integration / workflow automation support

B3. Illustrative provider examples

The examples below are illustrative only and are grouped by function. They are not recommendations, rankings, or evidence of suitability for this case.

B3.1 Legal & regulatory advisers

These advisers help determine whether the proposed structure is legally enforceable, properly classified, and consistent with the intended issuance, distribution, transfer, custody, and disclosure model.

Typical use cases:

- wrapper design
- securities-law classification

- cross-border marketing analysis
- transfer restriction design
- offering documentation
- legal opinions and regulatory scoping

Illustrative examples:

- CMS
- Cooley
- FIN LAW
- MME
- NÄGELE

B3.2 Issuing platforms

These providers typically support issuance workflows, lifecycle management, permissions, investor access rules, and parts of the reporting or compliance stack.

Typical use cases:

- token issuance
- lifecycle events
- permissions / whitelisting
- cap-table or investor-position management
- compliance rule integration
- reporting workflows

Illustrative examples:

- Aktionariat
- Polymesh (Polymath)
- Realiz Group
- Securitize
- Token City
- Tokeny

B3.3 Custody, trading & settlement

These providers support safekeeping, wallet infrastructure, execution, venue access, settlement, and post-trade operations.

Typical use cases:

- institutional custody
- secure wallet infrastructure
- settlement support
- exchange / venue access
- post-trade servicing
- collateral mobility or transfer control support

Illustrative examples:

- ADDX
- Anchorage Digital
- Archax
- BitGo
- Coinbase Institutional

- Copper
- Ondo Finance
- SIX Digital Exchange (SDX)
- Sygnum Bank
- Taurus

B4. Practical selection principles

When evaluating providers, the reader should typically assess:

- jurisdictional fit
- regulatory status where relevant
- experience with the relevant asset class
- operating-model compatibility
- integration burden
- exit options / portability
- fee model
- service resilience
- auditability and evidence quality
- concentration risk

Where possible, provider choices should reinforce the case logic rather than undermine it. A weak provider stack can easily erode the value thesis of an otherwise promising structure.

B5. Important caution

The right provider stack is highly case-dependent.

A provider that is suitable for a closed institutional collateral workflow may be a poor fit for a cross-border fund wrapper. A provider that is strong in digital-asset custody may not be the right choice for transfer administration or legal structuring. A venue that looks attractive commercially may create regulatory or operating-model problems if the case does not actually need that level of transferability.

Appendix D — Glossary & abbreviations

A practical reference for the most common terms and abbreviations used in this report. Spelling note: tokenization uses “z”.

Term	Meaning
Accredited investor	An investor category used primarily in the United States for persons or entities meeting specific income, net worth, or institutional eligibility thresholds for participation in certain exempt offerings.
Added Value (AV)	TVAC’s net assessment of whether tokenization appears to improve a case overall once both upside and friction are weighed together. Formula: $AV = (NO + CS + RR) - (TC + NR)$.
Asset owner / Issuer	The originating party, sponsor, manager, or issuing structure responsible for launching or sponsoring the tokenized case being assessed.
ATS	Alternative Trading System. A regulated trading venue category used in some jurisdictions, particularly the United States.
Cash leg	The cash side of a transaction or settlement flow, as distinct from the asset or token side.
Conditional Go	A Tokenization Value Verdict indicating that tokenization may add net value, but remains materially dependent on design discipline, implementation choices, and resolution of key gating conditions.
Custody	The safeguarding, control, and recordkeeping of assets or positions on behalf of investors or market participants.
Distribution scope	The intended investor reach of an offering or tokenized structure, including geography, investor type, and access conditions.
DvP	Delivery versus Payment. A settlement model in which transfer of the asset and transfer of payment are linked so that one does not occur without the other.
Economic rights	The financial entitlements attached to a token or instrument, such as income participation, coupon, principal repayment, or profit-sharing.
Execution risk	The risk that a concept that appears viable in theory fails in practice because implementation, coordination, economics, or governance proves weaker than assumed.
Governance	The decision-making, control, accountability, and escalation framework surrounding a structure, platform, issuer, or operating model.
Institutional investor	A professional market participant such as a fund, bank, insurer, treasury function, pension investor, or other organization investing at scale.
Investor segmentation	The way the target investor base is defined and limited by investor type, sophistication, jurisdiction, or eligibility criteria.
Net Added Value	See Added Value (AV).
New Opportunities (NO)	Case-specific upside that tokenization creates or materially strengthens, such as new investor reach, new operational utility, collateral use, programmability, or improved strategic flexibility.
New Risks (NR)	Additional risks introduced by tokenization, including legal, regulatory, operational, technological, governance, distribution, or market-structure risks.
NO / CS / RR / TC / NR	Shorthand for TVAC’s five factors: New Opportunities, Cost Savings, Risk Reduction, Tokenization Costs, and New Risks.

Term	Meaning
Non-viable	A Tokenization Value Verdict indicating that the current structure does not support a positive tokenization value case under the TVAC methodology and should not proceed without substantial redesign.
Parallel processes	A situation in which tokenized and legacy workflows both remain active, limiting the real efficiency gains from tokenization.
Permissioned	A system design in which access, participation, or transfers are restricted to approved participants under defined rules.
RACI	Responsible, Accountable, Consulted, Informed. A responsibility matrix used to make clear who performs, owns, advises on and must be kept informed about a task or control.
Registry	The official or operational record of holdings, ownership, or investor positions.
Repo	Repurchase agreement. A financing transaction in which securities are sold and later repurchased, typically functioning as a collateralized short-term funding arrangement.
Retail investor	A non-professional investor who generally receives higher levels of regulatory protection than professional or institutional investors.
Rights mapping	The process of making clear what legal, economic, governance, and transfer rights a token actually represents and how those rights are enforceable.
Risk Reduction (RR)	The extent to which tokenization reduces existing risk in the conventional setup, for example through better controls, improved transparency, clearer auditability, or lower reconciliation burden.
SEC	Securities and Exchange Commission in the United States.
Settlement model	The way asset transfer, cash transfer, record updates, and finality are operationally achieved.
SPV	Special Purpose Vehicle. A ring-fenced legal entity often used for issuance, structuring, asset isolation, or liability segregation.
Tokenization Costs (TC)	The additional costs introduced by tokenization, including structuring, legal work, compliance, integration, platform costs, governance overhead, servicing changes, and operating complexity.
Transfer controls	Rules, permissions, and technical or legal mechanisms governing who may transfer, receive, or hold the token.
Transfer restrictions	Specific limitations placed on secondary transfers, often for regulatory, contractual, or investor-eligibility reasons.
TVAC	Tokenization Value-Add Calculator. A structured decision-support methodology for assessing whether tokenization appears to create net added value in a specific case.
Vendor-capture risk	The risk that a structure becomes economically or operationally overly dependent on one platform, provider, or implementation path in a way that shifts value away from the initiating party.
Viable	A Tokenization Value Verdict indicating that the TVAC methodology identifies a positive tokenization value case under the described structure, subject to validation and disciplined execution. It is not an investment, credit, yield, valuation, legal, tax, or suitability assessment.

Term	Meaning
Whitelist / Whitelisting	A mechanism for restricting participation or transferability to pre-approved investors, wallets, counterparties, or participants.

Appendix E — User-supplied case inputs

E.1 Full verbatim case description

Full original free-text case input provided by the user.

We are a US-based corporate issuer with an existing, recurring commercial paper (CP) program used for short-term working capital. Today the program relies on multiple intermediaries, manual confirmations, repeated reconciliations between the issuer's treasury team, issuing/placing agents, custodians, and investors, and a settlement cycle that creates operational friction and occasional settlement fails.

We want to introduce a tokenized CP note format (economically equivalent to our existing CP) for professional and accredited investors only. This is not a retail product and not a public exchange initiative. The scope is deliberately conservative: a single US issuance + distribution perimeter, whitelisted participation, and institutional-grade custody. The primary objective is post-trade efficiency and risk reduction rather than “new liquidity”.

The tokenized format is designed to (1) reduce reconciliation breaks by moving to a single shared position record, (2) automate issuance and lifecycle events (issuance confirmations, coupon/accrual reporting, maturity repayment notices), and (3) tighten settlement controls through delivery-versus-payment (DvP) style workflows. Importantly, we commit to decommissioning duplicated legacy steps for the tokenized tranche (no permanent parallel “double-booking”), so savings are structural rather than cosmetic.

We expect measurable cost savings (reduced agent/custody operations per issuance, fewer manual exceptions) and measurable risk reduction (fewer settlement fails, cleaner audit trail, reduced counterparty/operational exposure). Any secondary transfers, if permitted at all, are tightly controlled and limited to eligible whitelisted holders under standard securities restrictions. The issuer is the primary beneficiary of the operational savings and control improvements under this design.

E.2 Structured input record

This record reproduces the structured inputs supplied for this assessment. For dropdowns and checkboxes, it also shows the alternatives that were available but not selected.

Field	User response	Other available options not selected
Issuance jurisdiction	United States	Not applicable
Sale jurisdictions (comma-separated)	United States	Not applicable
Instrument type	Digital wrapper (CSD/TA/registrar)	Sole-record / native tokenized security
Custody model	Regulated custodian	Transfer agent / CSD; Self-custody (wallets)
Secondary-market liquidity	None	Ad-hoc OTC; Quarterly windows; Continuous
Secondary-market venue	OTC	ATS / MTF; DEX (permissioned); CSD / TA-registered
Payment / cash-leg model	T+1/T+2 securities settlement	Not decided yet; Not material at this stage; Traditional bank / fiat payments; Custodian / bank / TA / CSD settlement; Stablecoin / tokenized cash; Deposit token / tokenized deposit / institutional DvP rail; Advanced: T+0 / atomic DvP; Advanced: off-chain booking
Offering disclosure	Exempt / limited-offering route (jurisdiction-dependent)	KID (PRIIPs); Prospectus
Transfer restrictions	Whitelist + accredited/professional-only + issuer/agent-approved transfers	Not applicable
Tax considerations (optional)	Standard US withholding and investor reporting depend on holder type; tax handled under existing CP program processes.	Not applicable
Investor types	Professional; Qualified / accredited (jurisdiction-dependent)	Retail

An option shown as not selected was not necessarily actively considered or rejected by the user; it was simply not selected for this assessment.

Scenario comparison: Alternative options shown here can help identify design variables worth testing in a new assessment or in TVAC Scenario Lab—for example a different custody model, settlement model, participant type or transfer setup.

Appendix F — Disclaimer

Disclaimer

This report is provided for informational and decision-support purposes only. It does not constitute legal, regulatory, tax, accounting, investment, valuation, or financial advice, and it should not be relied upon as a substitute for professional advice tailored to the specific facts and circumstances of the case.

TVAC is a structured assessment tool designed to evaluate whether and when tokenization may create added value relative to conventional alternatives. Its outputs reflect a methodology-based analytical framework operating under uncertainty. They are not guarantees of commercial success, feasibility, investor demand, regulatory approval, liquidity, execution quality, budget outcome, or risk reduction.

Any scores, verdicts, scenarios, stakeholder assessments, suggested actions, implementation observations, and roadmap elements in this report are indicative only and depend on the quality, completeness, and interpretation of the input data, as well as on assumptions about structure, jurisdiction, service-provider readiness, market conditions, execution capability, and timing.

Tokenization outcomes can differ materially from this assessment if key conditions change, including legal structuring, investor scope, jurisdiction count, custody model, distribution pathway, settlement design, governance arrangements, servicing readiness, technology integration, cost allocation, evidence quality, or liquidity support.

A TVAC verdict is a tokenization added-value verdict only. It must not be presented or used as investment approval, credit rating, yield validation, issuer endorsement, regulatory approval, due diligence report, or investor recommendation. TVAC does not validate projected returns, coupon payments, credit quality, valuations, legal enforceability, tax treatment, or investor suitability.

This report does not replace formal legal analysis, regulatory review, tax analysis, technical architecture review, cybersecurity review, accounting assessment, operational diligence, commercial validation, or board-level decision processes. Independent advice should always be obtained before any structuring, issuance, distribution, investment, operational, or implementation decision is made.

Neither TVAC nor Astoval assumes responsibility for actions taken on the basis of this report alone. Users remain solely responsible for all commercial, legal, regulatory, operational, and strategic decisions related to any proposed tokenization initiative.



Tokenization Value-Add Calculator

A PRODUCT BY ASTOVAL

Astoval provides independent, AI-enhanced early-stage decision support for asset tokenization, helping organizations assess whether tokenization creates genuine added value before significant resources are committed to legal structuring, platform selection or implementation.

PRIVACY AND DATA

Case inputs and generated reports are processed for delivery and are not intentionally retained as a persistent customer-case record in the standard self-service flow. OpenAI API data is not used for model training unless the API customer opts in; separate abuse-monitoring retention may apply for up to 30 days.

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